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Music, Mind & Artificial Intelligence

Assignment 1

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While I listen to a wide variety of genres, the music that resonates with me most deeply is Cantopop. One song in particular "青春頌 (Youth Anthem)" produced by Alfred Hui (許廷鏗) has followed me across three distinct periods of my life, each symbolising a significant growth in my teenagehood. Not only does it encapsulate a coming-of-age anthem, but it also serves as a case study on how the tonal structure of Cantonese shapes melodic composition, how episodic memory binds music to lived experience, and how a song's meaning transforms as the listener does.

Being born and raised in Hong Kong, I found that Alfred Hui's background and career path has always resonated with me. In fact, he trained as a dentist before winning a TVB singing competition and pivoting to music full time, which reminded me artistic ambition and professional training are not mutually exclusive. Best known for emotional ballads, "Youth Anthem" is among his most widely recognized tracks. Musically, "Youth Anthem" is a piano driven ballad in C major at approximately 139 beats per minute, adapting a verse chorus bridge structure, opening with a sparse piano accompaniment beneath Hui's vocal line before entering into a full arrangement in the chorus with layered strings and percussion. The lyrics address youth not as a carefree period but as something inherently messy, an era defined by uncertainty, struggle, and the slow process of learning to face difficulty alone. For almost any Hong Kong high schooler, the sentiment is immediately recognizable.

As a native Cantonese speaker, while I didn't have to learn that Cantonese is a tonal language, I had to realize it. I found out that Cantonese has six tones, with some references identifying up to nine. The pitch contour of a syllable determines its lexical meaning, which is the same sequence of consonants and vowels can refer to entirely different words depending on whether the pitch rises, falls, or remains level. By contrast, this is fundamentally different from English, where pitch conveys emotion or emphasis but never changes the meaning of a word.

Hence, such distinction has direct effects for songwriting. In Cantopop, melody must follow the tonal contours of the lyrics. For instance, a rising tone syllable set to a falling melodic line risks becoming unintelligible or, worse, changing the word's meaning entirely. Therefore, Cantopop composers face a constraint that English language songwriters simply do not: the melody is partially dictated by the words themselves.

Besson et al. (2017) argue that music and language both include the fundamental acoustic parameters such as frequency, duration, timbre, and so on and propose a "cascade" model where training in one domain would transfer to the other. Patel (2008) further supports this convergence, highlighting that music and language rely on shared syntactic integration resources in frontal brain regions, meaning that for a tonal languages such as Cantonese, the brain isn't just serving as a function to process melody and lyrics side by side but directing them through overlapping neural pathways at the same time. As a Cantonese listener, the linguistic pitch contours of the lyrics and the musical pitch of the melody equally reinforce each other at once. For example, when the melody goes along with the tonal contour, the song would feel natural and the lyrics land with immediate clarity. Besson et al. support this with research on Thai, another tonal language, implying that musically sensitive listeners learn tonal words more efficiently, implying that melodic and linguistic pitch processing share neural resources.

This idea eventually extends to brain structure, namely, Broca's area, traditionally associated with language syntax, also activates during musical syntactic processing (Besson et al., 2017). In Cantonese music, this kind of convergence is especially tight because pitch carries both linguistic and musical meaning at the same time. According to Deutsch et al. (2004), Mandarin speaking conservatory students displayed approximately 75% rates of absolute pitch, compared to dramatically lower rates among English speakers, suggesting that tonal language exposure trains the brain to encode pitch meaning associations from infancy. Cantonese, with more tonal categories than Mandarin, may potentially produce a stronger version of this effect. In "Youth Anthem," this constraint is audible. Consider the following excerpt:

"誰亦會隨年變老 方懂得童年多好 等不到來年變老 才懷念 那種悠然腳步
來吧趁毫無控訴 快釋穿沿途的污糟 這段少年時間 既然昂貴
大好青春要盡耗"

In this excerpt, the melodic contour consistently mirrors the tonal demands of the lyrics. For instance, the phrase 既然昂貴 rises in melodic pitch, aligning with the tonal movement from 昂 (ngong4, a low falling tone) to 貴 (gwai3, a mid level tone), where the natural Cantonese pronunciation already moves upward in register. Similarly, the phrase ending on 污糟 (wu1 zou1, both high level tones) descends melodically, for which the composer can replicate because the two syllables share the same tone category and do not risk misinterpretation. Meanwhile, the upward melodic sweep on 大好青春要盡耗 carries the emotional climax of the verse, and because the key syllables don't fight against their tonal contours, the lyrical meaning lands with full clarity even as the melody intensifies.

I first encountered "Youth Anthem" during the summer of 2023, at a Rites of Passage program in Australia, a mandatory leadership training for sophomores. It was my first long time away from home, and it was also where I experienced bullying for the first time. Other students talked behind my back, mocking how slowly I walked during steep hill treks, mistaking physical struggle for weakness. During those weeks, I kept replaying one line about facing storms alone and reframed what I was going through: struggling was part of the deal, not a sign that I was failing. Then, at the closing ceremony, my group performed the song together and what was once a private source of comfort transformed into a shared expression.

Two years later, in the summer of 2025, I sang the song again, this time at a karaoke session with friends I had met during a post exam internship. The emotional context could not have been more different. There was no isolation, no bullying. There was relief after the national exams, camaraderie, and the easy warmth of new friendships. The same melody carried an entirely different meaning. Meyer's theory of musical emotion, as discussed by Lee (2017), argues that emotional response to music arises from expectation.

In 2023, I expected the song to provide comfort in the face of difficulty. In 2025, I expected it to mark a shared celebration. The song's acoustic content did not change between those two summers; my expectations did. That shift in expectation is what shifted the meaning. Large (2017) defines music as "an interaction between what we hear and what we know." Just like what I knew about youth, about struggle, about belonging, had largely changed, the song also changed with it.

Now, writing this paper, hearing even the opening piano figure activates both memories at once. The 2025 karaoke did not erase the 2023 experience. It layered new associations on top of the original trace. The song now carries both pain and joy of growth, a bittersweet feeling, not as contradiction, but as accumulated meaning.

Hui's catalog is dominated by love ballads and vocal showcase tracks. Songs like "護航" and "面具" are technically impressive but thematically centered on romance. "Youth Anthem" differs in two aspects. First, its theme, struggles and youth, is more universally relatable than romantic heartbreak. Second, the arrangement is more restrained than Hui's typical work, where the emotional weight comes from the lyrics and their context rather than vocal pyrotechnics.

However, the deepest difference is personal and this one is the music I have lived through. It has been encoded, re-encoded, and retrieved across three distinct stages in my life, each encounter multiplying a layer of meaning the previous one did not have. "Youth Anthem" operates on two levels: as a Cantopop ballad shaped by the tonal constraints of Cantonese, and as an emotional artifact that has accumulated meaning across multiple re-encodings. Besson et al.'s (2017) cascade model illustrates why the melody and lyrics seem to be inseparable, where they are processed through shared acoustic and neural pathways. Patel's (2008) shared syntactic integration hypothesis explains why this processing is especially tight in a tonal language, where pitch carries both linguistic grammar and musical structure. Meyer's theory of expectation, as outlined in Lee (2017), explains why the same song meant survival in one summer and celebration in another.

The above redefines music as an interaction between what we hear and what we know (Large, 2017). What I knew changed across those three summers. The song's frequencies stayed the same, but its meaning did not.

Cited Sources

Besson, M., Barbaroux, M., & Dittinger, E. (2017). Music in the brain: Music and language processing. In R. Ashley & R. Timmers (Eds.), *The Routledge companion to music cognition* (pp. 37–48). Routledge.

Deutsch, D., Henthorn, T., & Dolson, M. (2004). Absolute pitch, speech, and tone language: Some experiments and a proposed framework. *Music Perception*, *21*(3), 339–356.

Large, E. W. (2017). Music from the air to the brain and body. In R. Ashley & R. Timmers (Eds.), *The Routledge companion to music cognition* (pp. 3–12). Routledge.

Lee, K. M. (2017). Music cognition: Investigations through the centuries. In R. Ashley & R. Timmers (Eds.), *The Routledge companion to music cognition* (pp. 467–478). Routledge.

Patel, A. D. (2008). *Music, language, and the brain*. Oxford University Press.